Sriyank Sagi

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EDUCATION

University of Maryland, College Park, Maryland
Master of Science in Data Science (GPA: 4.0/4.0) | August 2024 – May 2026 (Expected)
Amrita Vishwa Vidyapeetham, Coimbatore, India
Bachelor of Technology in Electrical and Computer Engineering | 2020 – 2024

PROFESSIONAL EXPERIENCE

Data Science Researcher | January 2025 – Present

- Collected and processed large-scale customer transaction data using Spark Delta tables, enabling efficient storage, retrieval, and analysis of spending patterns for fraud detection and financial insights.
- Implemented advanced NLP techniques, leveraging VADER and BERT for sentiment analysis of customer reviews and chatbot interactions. Integrated a RAG sequence model for automated customer support response generation, improving user engagement and satisfaction.

Data Scientist Intern | January 2024 – June 2024
Magnasoft, Hyderabad, India

- Trained a YOLOv8 model on 100,000+ aerial images to detect and classify objects with high precision for aerial imagery analysis, leveraging GeoTIFF data for geospatial insights.
- Built an interactive dashboard integrating OSRM-based real-time tracking, enabling object visualization, geolocation mapping, and dynamic analytics for aerial surveillance.
- Developed a Gait-Based Deep Learning Model for human identification and anomaly detection, preprocessing video frames by extracting motion sequences and segmenting them into spatiotemporal patches for enhanced feature learning, trained on CASIA-B and OU-ISIR datasets with over 500,000 samples.
- Integrated a real-time surveillance system using OpenCV and TensorRT for accelerated inference, achieving a 93.8% accuracy in distinguishing normal vs. suspicious behavior in live monitoring scenarios.

Data Analysis Intern (Freelancer) | September 2022 - December 2022
Self-Employed, Hyderabad, India

- Analysed 120,000+ financial transactions using Pandas and SQL, applying statistical methods and anomaly detection techniques to identify spending patterns, income trends, and fraud risks.
- Developed interactive visualizations using Matplotlib, Seaborn, and Plotly, including time-series analysis, heat maps, and geographic distribution plots to uncover actionable insights for financial decision-making.

PROJECTS

Market Power Price Prediction|python

- Developed a machine learning model using Long Short-Term Memory (LSTM) networks for time-series forecasting and Genetic Algorithms for feature selection and hyperparameter optimization, trained on over 113,000 real-time market data samples from the Indian Energy Exchange (IEX) company.
- Delivered precise market price predictions with improved forecasting accuracy, enabling enhanced decision-making for manufacturers and market participants by leveraging Python, Pandas and SciPy to capture temporal trends, optimize model performance, and ensure scalable, real time forecasting capabilities.

Fraud Detection for Financial Transactions | Python, C, Random Forest, Autoencoders, MySQL, Power BI

- Data Analysis & Model Development: Analyzed financial transaction records using Pandas and MySQL to detect anomalies. Built Random Forest and Autoencoder models for fraud detection, and optimized data retrieval with a C pipeline.
- Data Visualization & Insights: Created an interactive Power BI dashboard to visualize transaction flows and highlight suspicious patterns. Used MySQL for querying large datasets and optimizing performance.

Learning Management System (LMS) | React, Django, PostgreSQL, Firebase

- Full-Stack Development & API Integration: Designed an interactive LMS using Next.js and React.js for the frontend, with Django REST Framework for backend APIs. Implemented PostgreSQL for structured data storage and Firebase Firestore for real-time progress tracking. Developed features like video lectures, quizzes, progress tracking, and role-based authentication (students, instructors, admins).
- Scalability & Performance Optimization: Integrated WebSockets (Node.js) for real-time discussions and quizzes. Used AWS S3 Storage for efficient video streaming. Optimized database queries for fast course retrieval and auto-generated course completion certificates.

Hate Speech Detection using Natural Language Processing (NLP) | Python

- Automated the classification of social media comments as hateful or non-hateful by processing over 50,000 text samples using DistilBERT for feature extraction and Long Short-Term Memory (LSTM) networks for text classification, achieving an accuracy of 95% in detecting hate speech.
- • Developed a real-time web application using React.js for the frontend, Express.js and Node.js for the backend, with MongoDB as the database to store and manage user comments, enabling efficient content moderation and reducing harmful online interactions

TECHNICAL SKILLS

- **Languages**: C, C++, Python, HTML, CSS, JavaScript, Bash
- **Developer Tools:** Git, VS Code, PyCharm, Google Colab, Docker, Kubernetes,PowerBI
- **Frameworks**: PyTorch, Express, Node.js, Hadoop, Spark
- **Libraries**: scikit-learn, React.js, nltk, Bootstrap, Hugging Face Transformer, TensorFlow
- **Databases:** MongoDB, MySQL

CERTIFICATIONS

- **Google Data Analytics Professional Certificate|Google|2024**
- **IT automation with python certificate|Google|2024**
- **IBM Data science Professional certificate|IBM|2024**